[Aim: 100/100 in Maths]

आरंभ

QUADRATIC EQUATIONS

Lecture #2

#ATK²:-

Quadratic Equations

$\boxed{D = 2}$

=

eg $2x^2 - 3x + 4 = 0$

$x^2 = 0$

$x^2 + 4 = 0$

Zeroes | Roots | Solution

of Q.E

Methods of finding Solution of a Quadratic Equation:

① Splitting the Middle Term Method

or

FACTORISATION METHOD

② Discriminant Method

or

Quadratic Formula

or

श्री धराचार्य Method

#LP: Solve the following quadratic equation by factorisation:

(i) $x^2 - 17x + 60 = 0$

$p+q=-17$

$pq = 60$

$x^2 - 12x - 5x + 60 = 0$

$x(x-12) - 5(x-12) = 0$

$(x-12)[x-5] = 0$

2	60
2	30
3	15
5	5
	1

12

5

$x-12=0$ | $x-5=0$

$x=12$ | $x=5$

$x = 12, 5$

(ii) $x^2 - 3x - 40 = 0$

$p+q=-3$

$pq=-40$

$x^2 - 8x + 5x - 40 = 0$

$x(x-8) + 5(x-8) = 0$

$(x-8)[x+5] = 0$

2	40
2	20
2	10
5	5
	1

8

5

$x-8=0$ | $x+5=0$

$x=8$ | $x=-5$

$\boxed{x = 8, -5}$

(iii) $x^2 \underbrace{-13x} + 36 = 0$

$p+q = -13$

$pq = 36$

2	36
2	18
3	9
3	3
	1

4. 9.

$\Rightarrow \underbrace{x^2 - 9x} - \underbrace{4x + 36} = 0$

$\Rightarrow x(x-9) - 4(x-9) = 0$

$(x-9)[x-4] = 0$

$x - 9 = 0 \quad | \quad x - 4 = 0$

$x = 9 \quad | \quad x = 4$

(iv) $x^2 \underbrace{+10x} + 16 = 0$

$p+q = 10$

$pq = 16$

$\Rightarrow \underbrace{x^2 + 8x} + \underbrace{2x + 16} = 0$

$\Rightarrow x(x+8) + 2(x+8) = 0$

$\Rightarrow (x+8)[x+2] = 0$

$\Rightarrow x+8=0 \quad | \quad x+2=0$

$x = -8 \quad | \quad x = -2$

(v) $2x^2 + 1x - 300 = 0$

$\Rightarrow 2x^2 - 24x + 25x - 300 = 0$

$\Rightarrow 2x(x - 12) + 25(x - 12) = 0$

$\Rightarrow (x-12)(2x+25) = 0$

$x - 12 = 0$ | $2x + 25 = 0$

$x = 12$ | $x = -25/2$

$p + q = 1$

$pq = -600$

2	600
2	300
2	150
3	75
5	25
5	5
	1

24

25

(vi) $\sqrt{3}x^2 + 10x + 7\sqrt{3} = 0$

$\Rightarrow \sqrt{3}x^2 + 7x + 3x + 7\sqrt{3} = 0$

$x(\sqrt{3}x + 7) + \sqrt{3}(\sqrt{3}x + 7) = 0$

$(\sqrt{3}x + 7)[x + \sqrt{3}] = 0$

$\sqrt{3}x + 7 = 0$ | $x + \sqrt{3} = 0$

$\sqrt{3}x = -7$ | $x = -\sqrt{3}$

$x = \frac{-7}{\sqrt{3}}$

$p + q = 10$

$pq = 21$

$3, 7$

$p + q = 2\sqrt{2}$

$pq = -6$

(vii) $x^2 + 2\sqrt{2}x - 6 = 0$

$\Rightarrow x^2 + 3\sqrt{2}x - \sqrt{2}x - 6 = 0$

$\Rightarrow x(x + 3\sqrt{2}) - \sqrt{2}(x + 3\sqrt{2}) = 0$

$\Rightarrow (x + 3\sqrt{2})[x - \sqrt{2}] = 0$

$\Rightarrow x + 3\sqrt{2} = 0$ | $x - \sqrt{2} = 0$

$x = -3\sqrt{2}$ | $x = \sqrt{2}$

#LP: $4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 0$

(CBSE 2006,2013,2016)

$\Rightarrow \underbrace{4\sqrt{3}x^2 + 8x} - 3x - 2\sqrt{3} = 0$

$\Rightarrow \underline{4x}(\sqrt{3}x+2) \underline{-\sqrt{3}}(\sqrt{3}x+2) = 0$

$(\sqrt{3}x+2)[4x-\sqrt{3}] = 0$

$\sqrt{3}x+2=0$ | $4x-\sqrt{3}=0$

$x = -2/\sqrt{3}$ | $x = \sqrt{3}/4$

$p+q = 5$

$pq = 8\times 3$

$\Rightarrow 24$

#LP: $\dfrac{1}{x-1} - \dfrac{1}{x+5} = \dfrac{6}{7}, x \neq 1, -5$ **(CBSE 2010)**

LCM

$\Rightarrow \dfrac{1(x+5) - 1(x-1)}{(x-1)(x+5)} = \dfrac{6}{7}$

$\Rightarrow \dfrac{\cancel{x}+5 - \cancel{x}+1}{x^2+5x-x-5} = \dfrac{6}{7}$

$\Rightarrow \dfrac{\overset{1}{\cancel{6}}}{x^2+4x-5} = \dfrac{\overset{1}{\cancel{6}}}{7}$

$\Rightarrow 7 = x^2+4x-5$

$\Rightarrow 0 = x^2+4x-5-7$

$\Rightarrow$

$$\boxed{x^2+4x-12=0}$$

$x^2+6x-2x-12=0$

$x(x+6)-2(x+6)=0$

$(x+6)[x-2]=0$

$x+6=0$	$x-2=0$
$x=-6$	$x=2$

#LP: $\dfrac{x+3}{x+2} = \dfrac{3x-7}{2x-3}$, $x \neq -2, 3/2$ **(CBSE 2017)**

$\Rightarrow (x+3)(2x-3) = (3x-7)(x+2)$

$\Rightarrow \underbrace{2x^2} \underbrace{-3x} + \cancel{6x} \underbrace{-9} = \underbrace{3x^2} + \cancel{6x} \underbrace{-7x} \underbrace{-14}$

$\Rightarrow 0 = \underbrace{3x^2 - 2x^2} \underbrace{-7x + 3x} \underbrace{-14 + 9}$

$\Rightarrow \boxed{x^2 - 4x - 5 = 0}$

$-5x + x$


PAGLU
दैनिक जागरण
RANK 1
500/500
THANK YOU !
They doubted. I DIDN'T.
Kya karne aaya hai ?
Jaake padhai kar !!!
BY SHRUTI SETHI AARAMBH BATCH